Risk Assessment Details #4193: GM2 Technology for the Poorest Billion - SOLARSAFE Bag Team

Please note: Where the original risk assessment text contains special characters (especially greater than or less than signs) the text may appear truncated. If in doubt please check in the app.

Also note that if any attachments are listed for this risk assessment they will need to be viewed within the App.

Supervisor:	Tom Bashford
Div:	W
Location:	Inglis Building: Dyson Centre; St. Catherine's College Grounds
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Description:	Engineering Tripos Part IIA Project, GM2: Technology for the Poorest Billion, 2025-26. SOLARSAFE project - Using hybrid solar drying for maize. Description of works: We plan to use the Dyson Centre and its equipment to build an enclosed solar dryer for maize using plastic bin bags, wooden structures, and hot glue. We will also procure a large water bladder (as a thermal storage system) , a small USB-powered electric fan, a few portable chargers, and some corn cobs. Description of experiments: We plan to run several multi-hour, possibly overnight, experiments on the efficacy of our modifications to the drying system on St. Catherine's College grounds (e.g. grass patch, parking lot). We are in the midst of sorting out approvals from staff from St. Catherine's to run these experiments.
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Emergency Instructions & First Aid:	Location of first aid kit: Dyson Centre front desk. Look for technicians/porters. Fellow students. SAFETY OFFICE NOTE: To contact a first aider, whilst in the CUED, please call 32766 from an internal phone or 01223 332766 from a mobile.
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Any Special Monitoring:	
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Further Control Measures:	
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Out of hours/Lone Working:	Ideally work in 3's (one to look after the one with the accident, and one to show the ambulance crew up to the Dyson Centre)
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PPE:	Eye goggles and gloves for use in Dyson Centre.

Hazards

Hazard	Effect - who might be harmed and how	Control Measures	Residual Risk
Moving parts on small fan	Entanglement	Loose clothing, jewellery and long hair to be kept clear of moving parts.	Likelihood 3 x Severity 1 = 3 (Low)
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Hot glue dripping on skin	- Burns - Reaction leading to further injury	- Appropriate positioning of hands and glue. - Understanding of mechanism before doing difficult processes - PPE and mindfulness and ensuring no distractions or multitasking	Likelihood 3 x Severity 2 = 6 (Low)
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Plastic bag blocking air supply	Suffocation	Keep plastic bag away from head, work in pairs	Likelihood 1 x Severity 2 = 2 (Low)
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Splinters from working with wood	- Piercing of skin - Infection	Use gloves when handling visibly splintered wood.	Likelihood 3 x Severity 2 = 6 (Low)
-	-	-	-
Contact from cutting tool which has slipped	Cuts	- Use gloves when cutting - Place object being cut on appropriate cutting surface - Do not let body be in a slip plane or plane of cutting of a tool	Likelihood 3 x Severity 2 = 6 (Low)
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Hot equipment	Burns	Use gloves and evaluate temperature of parts sensibly before grabbing them.	Likelihood 1 x Severity 3 = 3 (Low)
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Fan overheats during experiment	Electrical fire/explosion	- Place fan under some shade (i.e. umbrella) while the experimental apparatus sits under the sun. - Conduct periodic checks in for daytime experiments (i.e. 2 hourly) to ensure all equipment is in order. - Station team members nearby for rapid response - Clearly demarcate experiment area	Likelihood 1 x Severity 4 = 4 (Low)
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Water bladder bursts, spilling contents onto floor and making it slilppery	Slips and falls	Place experiment aparatus on grass/away from walking paths.	Likelihood 2 x Severity 2 = 4 (Low)
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Inclement weather	Electronics left under rain might malfunction, leading to shocks or overheating.	- Place electronics under cover - Elevate electronics to avoid water on the ground - Monitor weather frequently - Station team members nearby for rapid response - Clearly demarcate experiment area	Likelihood 3 x Severity 3 = 9 (Medium)
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Attachments associated with this risk assessment (please check these using the app):
No attachments submitted